

Product Name: Zoledronic Acid 4mg/5ml	Territory: MY	Colour: ● BLACK ● DIE CUT	1. Draft 1. Corr	03.09.2019, 13:00 06.09.2019, 11:30	Variable Data:
Type of Packaging: PIL	Dosage: 1x5ml VIAL				
Material number: M090621/02 MY	2-D-Matrix Code X				
Pharma-Code (Logos): 2838	PACK-INSERT - INSTRUCTIONS FOR USE - READ CAREFULLY!				
Dimension: 180 x 882 mm	Smallest Size: 8 Pt.		Operator: Roberto Grill		

Zoledronic Acid Kabi, 4mg/5 mL Concentrate for Solution for Infusion

Composition

Each 5 mL of Zoledronic Acid Kabi 4mg/5ml Concentrate Solution contains:

Zoledronic acid 4 mg.

Pharmaceutical form

Concentrate for Solution for Infusion.

Zoledronic Acid Kabi 4 mg/5 mL Concentrate for Solution for Infusion is a clear, colorless, sterile solution, practically free from visible particles aqueous solution.

The reconstituted solution should be clear and free from visible particles.

Zoledronic Acid Kabi 4 mg/5 mL Concentrate for Solution for Infusion is filled into 10 mL polypropylene vials moulded from polypropylene resin. The vial is closed with halobutyl rubber stopper. It is sealed with a flip off caps made of aluminium with a plastic disc. The flip off caps do not come into contact with the finished product. The vial is packed in an outer carton made of cardboard.

Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion, further diluted into 100ml of calcium-free infusion solution (0.9% w/v sodium chloride or 5% w/v glucose solution), should be given as a single intravenous infusion

Therapeutic indications

- Prevention of skeletal related events (pathological fractures, spinal compression, radiation or surgery to bone, or tumour-induced hypercalcaemia) in patients with advanced malignancies involving bone.
- Treatment of hypercalcaemia of malignancy (HCM)

Posology and method of administration

Zoledronic Acid Kabi 4 mg/5ml Concentrate for Solution for Infusion must only be prescribed and administered to patients by healthcare professionals experienced in the administration of intravenous bisphosphonates.

Posology

Prevention of skeletal related events in patients with advanced malignancies involving bone

Adults and elderly

The recommended dose in the prevention of skeletal related events in patients with advanced malignancies involving bone is 4 mg zoledronic acid every 3 to 4 weeks.

Patients should also be administered an oral calcium supplement of 500 mg and 400 IU vitamin D daily. The decision to treat patients with bone metastases for the prevention of skeletal related events should consider that the onset of treatment effect is 2-3 months.

Treatment of HCM

Adults and elderly

The recommended dose in hypercalcaemia (albumin-corrected serum calcium ≥ 12.0 mg/dl or 3.0 mmol/l) is a single dose of 4 mg zoledronic acid.

Renal impairment HCM:

Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion treatment in patients with hypercalcaemia of malignancy (HCM) and who have severe renal impairment should be considered only after evaluating the risks and benefits of treatment. In the clinical studies, patients with serum creatinine $> 400 \mu\text{mol/l}$ or > 4.5 mg/dl were excluded. No dose adjustment is necessary in HCM patients with serum creatinine $< 400 \mu\text{mol/l}$ or < 4.5 mg/dl (see section **Warning and Precautions**).

Prevention of skeletal related events in patients with advanced malignancies involving bone:

When initiating treatment with Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion in patients with multiple myeloma or metastatic bone lesions from solid tumours, serum creatinine and creatinine clearance (CLCr) should be determined. CLCr is calculated from serum creatinine using the Cockcroft-Gault formula.

Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion is not recommended for patients presenting with severe renal impairment prior to initiation of therapy, which is defined for this population as CLCr < 30 mL/min. In clinical trials with zoledronic acid, patients with serum creatinine $> 265 \mu\text{mol/l}$ or > 3.0 mg/dl were excluded.

In patients with bone metastases presenting with mild to moderate renal impairment prior to initiation of therapy, which is defined for this population as CLCr $30\text{--}60$ mL/min, the following Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion dose is recommended (see also section **Warning and Precautions**):

Baseline Creatinine Clearance (mL/min)	Zoledronic acid Presenius Kabi Recommended Dose *
> 60	4.0 mg zoledronic acid
$50\text{--}60$	3.5 mg* zoledronic acid
$40\text{--}49$	3.3 mg* zoledronic acid
$30\text{--}39$	3.0 mg* zoledronic acid

* Doses have been calculated assuming target AUC of 0.66 (mg•h/L) (CLCr=75 mL/min). The reduced doses for patients with renal impairment are expected to achieve the same AUC as that seen in patients with creatinine clearance of 75 mL/min.

Following initiation of therapy, serum creatinine should be measured prior to each dose of zoledronic acid and treatment should be withheld if renal function has deteriorated. In the clinical trials, renal deterioration was defined as follows:

- For patients with normal baseline serum creatinine (< 1.4 mg/dl or $< 124 \mu\text{mol/l}$), an increase of 0.5 mg/dl or 44 $\mu\text{mol/l}$;
- For patients with an abnormal baseline creatinine (> 1.4 mg/dl or $> 124 \mu\text{mol/l}$), an increase of 1.0 mg/dl or 88 $\mu\text{mol/l}$.

In the clinical studies, zoledronic acid treatment was resumed only when the creatinine level returned to within 10% of the baseline value (see section **Warning and Precautions**). Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion treatment should be resumed at the same dose as that given prior to treatment interruption.

Paediatric population

The safety and efficacy of zoledronic acid in children aged 1 year to 17 years have not been established. Currently available data are described in sections **Warning and Precautions** and **Pharmacodynamics** but no recommendation on a posology can be made.

Method of administration

Intravenous use

Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion, further diluted in 100 mL, should be given as a single intravenous infusion in no less than 15 minutes.

In patients with mild to moderate renal impairment, reduced Zoledronic Acid Kabi 4mg/5ml concentrate for Solution for Infusion doses are recommended (see section **Posology** above and section **Warning and Precaution**).

Instructions for preparing reduced doses of Zoledronic Acid Kabi:

Withdraw an appropriate volume of the concentrate needed, as follows:

- 4.4 mL for 3.5 mg dose
- 4.1 mL for 3.3 mg dose
- 3.8 mL for 3.0 mg dose

The withdrawn amount of concentrate must be further diluted in 100 mL of sterile 0.9% w/v sodium chloride solution or 5% w/v glucose solution. The dose must be given as a single intravenous infusion over no less than 15 minutes.

Zoledronic Acid Kabi 4 mg/5ml Concentrate for Solution for Infusion must not be mixed with calcium or other divalent cation-containing infusion solutions such as lactated Ringer's solution, and should be administered as a single intravenous solution in a separate infusion line.

Patients must be maintained well hydrated prior to and following administration of Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion.

Contraindications

- Hypersensitivity to zoledronic acid or other bisphosphonates or any of the excipients in the formulation.
- Pregnancy and breast-feeding women.

Warning and Precautions

General

Patients must be assessed prior to administration of Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion to ensure that they are adequately hydrated. Overhydration should be avoided in patients at risk of cardiac failure.

Standard hypercalcaemia-related metabolic parameters, such as serum levels of calcium, phosphate and magnesium, should be carefully monitored after initiating Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion therapy. If hypocalcaemia, hypophosphataemia, or hypomagnesaemia occurs, short-term supplemental therapy may be necessary. Untreated hypercalcaemia patients generally have some degree of renal function impairment, therefore careful renal function monitoring should be considered.

Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion contains the same active substance as found in Aclasta (zoledronic acid). Patients being treated with Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion should not be treated with Aclasta or any other bisphosphonate concomitantly, since the combined effects of these agents are unknown.

Renal insufficiency

Patients with Tumour-Induced Hypercalcaemia (TIH) and evidence of deterioration in renal function should be appropriately evaluated with consideration given as to whether the potential benefit of treatment with Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion outweighs the possible risk.

The decision to treat patients with bone metastases for the prevention of skeletal related events should consider that the onset of treatment effect is 2-3 months.

Zoledronic Acid has been associated with reports of renal dysfunction. Factors that may increase the potential for deterioration in renal function include dehydration, pre-existing renal impairment, multiple cycles of Zoledronic Acid and other bisphosphonates as well as use of other nephrotoxic medicinal products. While the risk is reduced with a dose of 4 mg zoledronic acid administered over 15 minutes, deterioration in renal function may still occur. Renal deterioration, progression to renal failure and dialysis has been reported in patients after the initial dose or a single dose of 4 mg zoledronic acid. Increases in serum creatinine also occur in some patients with chronic administration of Zoledronic Acid at recommended doses for prevention of skeletal related events, although less frequently.

Patients should have their serum creatinine levels assessed prior to each dose of Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion. Upon initiation of treatment in patients with bone metastases with mild to moderate renal impairment, lower doses of zoledronic acid are recommended. In patients who show evidence of renal deterioration during treatment, Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion should be withheld. Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion should only be resumed when serum creatinine returns to within 10% of baseline. Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion treatment should be resumed at the same dose as that given prior to treatment interruption.

In view of the potential impact of zoledronic acid on renal function, the lack of clinical safety data in patients with severe renal impairment (in clinical trials defined as serum creatinine $\geq 400 \mu\text{mol/l}$ or ≥ 4.5 mg/dl for patients with TIH and $\geq 265 \mu\text{mol/l}$ or ≥ 3.0 mg/dl for patients with cancer and bone metastases, respectively) at baseline and only limited pharmacokinetic data in patients with severe renal impairment at baseline (creatinine clearance < 30 mL/min), the use of Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion is not recommended in patients with severe renal impairment.

Hepatic insufficiency

As only limited clinical data are available in patients with severe hepatic insufficiency, no specific recommendations can be given for this patient population.

Osteonecrosis of the jaw

Osteonecrosis of the jaw (ONJ) has been reported in patients, predominantly those with cancer, receiving treatment with medicinal products that inhibit bone resorption, such as Zoledronic Acid. Many of these patients were also receiving chemotherapy and corticosteroids. The majority of reported cases have been associated with dental procedures such as tooth extraction. Many had signs of local infection including osteomyelitis.

The following risk factors should be considered when evaluating an individual's risk of developing ONJ:

- Potency of the bisphosphonate (higher risk for highly potent compounds), route of administration (higher risk for parenteral administration) and cumulative dose
- Cancer, chemotherapy, radiotherapy, corticosteroids, smoking
- History of dental disease, poor oral hygiene, periodontal disease, invasive dental procedures and poorly fitting dentures.

A dental examination with appropriate preventive dentistry should be considered prior to treatment with bisphosphonates in patients with concomitant risk factors.

While on treatment, these patients should avoid invasive dental procedures if possible. For patients who develop osteonecrosis of the jaw while on bisphosphonate therapy, dental surgery may exacerbate the condition. For patients requiring dental procedures, there are no data available to suggest whether discontinuation of bisphosphonate treatment reduces the risk of osteonecrosis of the jaw. Clinical judgement of the treating physician should guide the management plan of each patient based on individual benefit/risk assessment.

Osteonecrosis of the external auditory canal

Osteonecrosis of the external auditory canal has been reported with bisphosphonates, mainly in association with long-term therapy. Possible risk factors for osteonecrosis of the external auditory canal include steroid use and chemotherapy and/or local risk factors such as infection or trauma. The possibility of osteonecrosis of the external auditory canal should be considered in patients receiving bisphosphonates who present with ear symptoms including chronic ear infections.

Osteonecrosis of other anatomical sites

Cases of osteonecrosis of other anatomical sites including the hip and femur have been reported predominantly in adult cancer patients treated with bisphosphonates, including Zoledronic.

Musculoskeletal pain

In post-marketing experience, severe and occasionally incapacitating bone, joint, and/or muscle pain have been reported in patients taking Zoledronic acid. However, such reports have been infrequent. The time to onset of symptoms varied from one day to several months after starting treatment. Most patients had relief of symptoms after stopping treatment. A subset had recurrence of symptoms when rechallenged with Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion or another bisphosphonate.

Atypical fractures of the femur

Atypical subtrochanteric and diaphyseal femoral fractures have been reported with bisphosphonate therapy, primarily in patients receiving long-term treatment for osteoporosis. These transverse or short oblique fractures can occur anywhere along the femur from just below the lesser trochanter to just above the supracondylar flare. These fractures occur after minimal or no trauma and some patients experience thigh or groin pain, often associated with imaging features of stress fractures, weeks to months before presenting with a completed femoral fracture. Fractures are often bilateral; therefore the contralateral femur should be examined in bisphosphonate-treated patients who have sustained a femoral shaft fracture. Poor healing of these fractures has also been reported. Discontinuation of bisphosphonate therapy in patients suspected to have an atypical femur fracture should be considered pending evaluation of the patient, based on an individual benefit risk assessment.

During bisphosphonate treatment patients should be advised to report any thigh, hip or groin pain and any patient presenting with such symptoms should be evaluated for an incomplete femur fracture.

Hypocalcaemia

Hypocalcaemia has been reported in patients treated with Zoledronic Acid. Cardiac arrhythmias and neurologic adverse events (including convulsions, hypoaesthesia and tetany) have been reported secondary to cases of severe hypocalcaemia. Cases of severe hypocalcaemia requiring hospitalisation have been reported. In some instances, the hypocalcaemia may be life-threatening. Caution is advised when Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion is administered with medicinal products known to cause hypocalcaemia, as they may have a synergistic effect resulting in severe hypocalcaemia (see section 4.5). Serum calcium should be measured and hypocalcaemia must be corrected before initiating Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion therapy. Patients should be adequately supplemented with calcium and vitamin D.

Interaction with Other Medicament

Caution is advised when bisphosphonates are administered with aminoglycosides, since both agents may have an additive effect, resulting in a lower serum calcium level for longer periods than required.

Caution is indicated when Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion is used with other potentially nephrotoxic medicinal products. Attention should also be paid to the possibility of hypomagnesaemia developing during treatment.

In multiple myeloma patients, the risk of renal dysfunction may be increased when Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion is used in combination with thalidomide.

Reports of ONJ have been received in patients treated with zoledronic acid (as indicated **and in Posology and Method of Administration**) concomitant anti-angiogenic medicinal products.

Fertility, pregnancy and lactation

Pregnancy

There are no adequate data on the use of Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion in pregnant women. Animal reproduction studies with zoledronic acid have shown reproductive toxicity. The potential risk for humans is unknown. Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion should not be used during pregnancy.

Breast-feeding

It is not known whether zoledronic acid is excreted into human milk. Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion is contraindicated in breast-feeding women. (see section **Contraindication**).

Fertility

Zoledronic acid was evaluated in rats for potential adverse effects on fertility of the parental and F1 generation. This resulted in exaggerated pharmacological effects considered to be related to the compound's inhibition of skeletal calcium metabolism, resulting in periparturient hypocalcaemia, a bisphosphonate class effect, dystocia and early termination of the study. Thus these results precluded determining a definitive effect of zoledronic acid on fertility in humans.

Effects on ability to drive and use machines

Adverse reactions, such as dizziness and somnolence, may have influence on the ability to drive or use machines, therefore caution should be exercised with the use of Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion along with driving and operating of machinery.

Undesirable effects

Summary of the safety profile

Within three days after Zoledronic Acid administration, an acute phase reaction has commonly been reported, with symptoms including bone pain, fever, fatigue, arthralgia, myalgia, rigors and arthritis with subsequent joint swelling; these symptoms usually resolve within a few days (see description of selected adverse reactions).

The following are the important identified risks with Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion in the approved indications: Renal function impairment, osteonecrosis of the jaw, acute phase reaction, hypocalcaemia, atrial fibrillation, anaphylaxis, interstitial lung disease. The frequencies for each of these identified risks are shown in Table 1.

Tabulated list of adverse reactions

The following adverse reactions, listed in **Table 1**, have been accumulated from clinical studies and post-marketing reports following predominantly chronic treatment with 4 mg zoledronic acid:

Blood and lymphatic system disorders	
Common:	Anaemia
Uncommon:	Thrombocytopenia, leukopenia
Rare:	Pancytopenia
Immune system disorders	
Uncommon:	Hypersensitivity reaction
Rare:	Angioneurotic oedema
Psychiatric disorders	
Uncommon:	Anxiety, sleep disturbance
Rare:	Confusion
Nervous system disorders	
Common:	Headache
Uncommon:	Dizziness, paraesthesia, dysgeusia, hypoaesthesia, hyperaesthesia, tremor, somnolence
Very rare:	Convulsions, hypoaesthesia and tetany (secondary to hypocalcaemia)
Eye disorders	
Common:	Conjunctivitis
Uncommon:	Blurred vision, scleritis and orbital inflammation
Rare:	Uveitis
Very rare:	Episcleritis
Cardiac disorders	
Uncommon:	Hypertension, hypotension, atrial fibrillation, hypotension leading to syncope or circulatory collapse
Rare:	Bradycardia, cardiac arrhythmia (secondary to hypocalcaemia)
Respiratory, thoracic and mediastinal disorders	
Uncommon:	Dyspnoea, cough, bronchoconstriction
Rare:	Interstitial lung disease
Gastrointestinal disorders	
Common:	Nausea, vomiting, decreased appetite
Uncommon:	Diarrhoea, constipation, abdominal pain, dyspepsia, stomatitis, dry mouth
Skin and subcutaneous tissue disorders	
Uncommon:	Pruritus, rash (including erythematous and macular rash), increased sweating
Musculoskeletal and connective tissue disorders	
Common:	Bone pain, myalgia, arthralgia, generalised pain
Uncommon:	Muscle spasms, osteonecrosis of the jaw
Very rare:	Osteonecrosis of the external auditory canal (bisphosphonates class adverse reaction)
Renal and urinary disorders	
Common:	Renal impairment
Uncommon:	Acute renal failure, haematuria, proteinuria
Rare:	Acquired Fanconi Syndrome
General disorders and administration site conditions	
Common:	Fever, flu-like syndrome (including fatigue, rigors, malaise and flushing)
Uncommon:	Asthenia, peripheral oedema, injection site reactions (including pain, irritation, swelling, induration), chest pain, weight increase, anaphylactic reaction/shock, urticaria
Rare:	Arthritis and joint swelling as a symptom of acute phase reaction
Investigations	
Very common:	Hypophosphataemia
Common:	Blood creatinine and blood urea increased, hypocalcaemia
Uncommon:	Hypomagnesaemia, hypokalaemia
Rare:	Hyperkalaemia, hypernatraemia

Description of selected adverse reactions

Renal function impairment

Zoledronic Acid has been associated with reports of renal dysfunction in clinical trials for the prevention of skeletal-related events in patients with advanced malignancies involving bone. Factors that may increase the potential for deterioration in renal function include dehydration, pre-existing renal impairment, multiple cycles of Zoledronic Acid or other bisphosphonates, as well as concomitant use of nephrotoxic medicinal products or using a shorter infusion time than currently recommended. Renal deterioration, progression to renal failure and dialysis has been reported in patients after the initial dose or a single dose of 4 mg zoledronic acid (see section **Warning and Precautions**).

Osteonecrosis

Cases of osteonecrosis (primarily of the jaws) but also of other anatomical sites including hip, femur and external auditory canal have been reported predominantly in cancer patients treated with bisphosphonates, including Zoledronic Acid. Many patients with osteonecrosis of the jaw had signs of local infection including osteomyelitis, and the majority of the reports refer to cancer patients following tooth extractions or other dental surgeries. Osteonecrosis of the jaws has multiple well documented risk factors including a diagnosis of cancer, concomitant therapies (e.g. chemotherapy, anti-angiogenic drugs, radiotherapy, corticosteroids) and co-morbid conditions (e.g. anaemia, coagulopathies, infection, pre-existing oral disease). Although causality has not been determined, it is prudent to avoid dental surgery as recovery may be prolonged (see section **Warning and Precautions**).

Acute phase reaction

This adverse drug reaction consists of a constellation of symptoms that includes fever, myalgia, headache, extremity pain, nausea, vomiting, diarrhoea arthralgia and arthritis with subsequent joint swelling. The onset time is ≤ 3 days post Zoledronic Acid, and the reaction is also referred to using the terms “flu-like” or “post-dose” symptoms.

Hypocalcaemia-related ADRs

Hypocalcaemia is an important identified risk with Zoledronic Acid in the approved indications. Based on the review of both clinical trial and post-marketing cases, there is sufficient evidence to support an association between Zoledronic Acid therapy, the reported event of hypocalcaemia, and the secondary development of cardiac arrhythmia. Furthermore, there is evidence of an association between hypocalcaemia and secondary neurological events reported in these cases including; convulsions, hypoaesthesia and tetany (see section **Warning and Precautions**).

Symptoms and Treatment of Overdose

Clinical experience with acute overdose of zoledronic acid is limited. The administration of doses up to 48 mg of zoledronic acid in error has been reported. Patients who have received doses higher than those recommended (see section **Posology and method of administration**) should be carefully monitored, since renal function impairment (including renal failure) and serum electrolyte (including calcium, phosphorus and magnesium) abnormalities have been observed. In the event of hypocalcaemia, calcium gluconate infusions should be administered as clinically indicated.

PHARMACOLOGICAL PROPERTIES

Pharmacotherapeutic group: Drugs for treatment of bone diseases, bisphosphonate.
ATC code: M05BA08

Zoledronic acid belongs to the class of bisphosphonates and acts primarily on bone. It is an inhibitor of osteoclastic bone resorption. The selective action of bisphosphonates on bone is based on their high affinity for mineralised bone, but the precise molecular mechanism leading to the inhibition of osteoclastic activity is still unclear.

In long-term animal studies, zoledronic acid inhibits bone resorption without adversely affecting the formation, mineralisation or mechanical properties of bone.

In addition to being a potent inhibitor of bone resorption, zoledronic acid also possesses several anti tumour properties that could contribute to its overall efficacy in the treatment of metastatic bone disease. The following properties have been demonstrated in preclinical studies:

- *In vivo*: Inhibition of osteoclastic bone resorption, which alters the bone marrow micro environment, making it less conducive to tumour cell growth, anti-angiogenic activity and anti-pain activity.
- *In vitro*: Inhibition of osteoblast proliferation, direct cytostatic and pro-apoptotic activity on tumour cells, synergistic cytostatic effect with other anti-cancer drugs, anti-adhesion/invasion activity.

Pharmacokinetic properties

Single and multiple 5- and 15-minute infusions of 2, 4, 8 and 16 mg zoledronic acid in 64 patients with bone metastases yielded the following pharmacokinetic data, which were found to be dose independent. After initiating the infusion of zoledronic acid, the plasma concentrations of zoledronic acid rapidly increased, achieving their peak at the end of the infusion period, followed by a rapid decline to < 10% of peak after 4 hours and < 1% of peak after 24 hours, with a subsequent prolonged period of very low concentrations not exceeding 0.1% of peak prior to the second infusion of zoledronic acid on day 28. Intravenously administered zoledronic acid is eliminated by a triphasic process: rapid biphasic disappearance from the systemic circulation, with half-lives of t½α 0.24 and t½β 1.87 hours, followed by a long elimination phase with a terminal elimination half-life of t½γ 146 hours. There was no accumulation of zoledronic acid in plasma after multiple doses given every 28 days. Zoledronic acid is not metabolised and is excreted unchanged via the kidney. Over the first 24 hours, 39 ± 16% of the administered dose is recovered in the urine, while the remainder is principally bound to bone tissue.

From the bone tissue it is released very slowly back into the systemic circulation and eliminated via the kidney. The total body clearance is 5.04 ± 2.5 l/h, independent of dose, and unaffected by gender, age, race, and body weight. Increasing the infusion time from 5 to 15 minutes caused a 30% decrease in zoledronic acid concentration at the end of the infusion, but had no effect on the area under the plasma concentration versus time curve. The inter patient variability in pharmacokinetic parameters for zoledronic acid was high, as seen with other bisphosphonates.

No pharmacokinetic data for zoledronic acid are available in patients with hypercalcaemia or in patients with hepatic insufficiency. Zoledronic acid does not inhibit human P450 enzymes *in vitro*, shows no biotransformation and in animal studies < 3% of the administered dose was recovered in the faeces, suggesting no relevant role of liver function in the pharmacokinetics of zoledronic acid.

The renal clearance of zoledronic acid was correlated with creatinine clearance, renal clearance representing 75 ± 33% of the creatinine clearance, which showed a mean of 84 ± 29 ml/min (range 22 to 143 ml/min) in the 64 cancer patients studied. Population analysis showed that for a patient with creatinine clearance of 20 ml/min (severe renal impairment), or 50 ml/min (moderate impairment), the corresponding predicted clearance of zoledronic acid would be 37% or 72%, respectively, of that of a patient showing creatinine clearance of 84 ml/min. Only limited pharmacokinetic data are available in patients with severe renal insufficiency (creatinine clearance < 30 ml/min).

Zoledronic acid shows no affinity for the cellular components of blood and plasma protein binding is low (approximately 56%) and independent of the concentration of zoledronic acid.

Special populations

Paediatric patients

Limited pharmacokinetic data in children with severe osteogenesis imperfecta suggest that zoledronic acid pharmacokinetics in children aged 3 to 17 years are similar to those in adults at a similar mg/kg dose level. Age, body weight, gender and creatinine clearance appear to have no effect on zoledronic acid systemic exposure.

PHARMACEUTICAL PARTICULARS

List of excipients

Mannitol
Sodium citrate
Water for injections

Incompatibilities

To avoid potential incompatibilities, Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion is to be diluted with 0.9% w/v sodium chloride solution or 5% w/v glucose solution.

Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion must not be mixed with calcium or other divalent cation-containing infusion solutions such as lactated Ringer's solution, and should be administered as a single intravenous solution in a separate infusion line.

Shelf life

24 months.

From a microbiological point of view, the diluted solution for infusion should be used immediately. If not used immediately, chemical and physical in-use stability has been demonstrated for 24 hours at 2°C- 8°C. The refrigerated solution should then be equilibrated to room temperature prior to administration.

Pack size

Packs containing 1, 4 or 10 vials.

Special precautions for disposal

Prior to administration, 5.0 ml concentrate from one vial or the volume of the concentrate withdrawn as required must be further diluted with 100 ml of calcium-free infusion solution (0.9% w/v sodium chloride solution or 5% w/v glucose solution).

Studies with glass bottles, as well as several containers types made from polyvinylchloride, polyethylene and polypropylene (prefilled with 0.9% w/v sodium chloride solution or 5% w/v glucose solution), showed no incompatibility with Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion.

Additional information on handling of Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion, including guidance on preparation of reduced doses, is provided in section **Posology and Method of Administration**.

Aseptic techniques must be followed during the preparation of the infusion.

For single use only.

Only clear solution free from particles and discoloration should be used.

Healthcare professionals are advised not to dispose of unused Zoledronic Acid Kabi 4mg/5ml Concentrate for Solution for Infusion via the domestic sewage system. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

MANUFACTURER:

Fresenius Kabi Austria GMBH
Hafnerstraße 36
8055 Graz, Austria

PRODUCT REGISTRATION HOLDER/IMPORTER:

Fresenius Kabi Malaysia Sdn Bhd #3-1 & 3-2, Axis Technology Centre, Lot 13, Jalan 51A/225, 46100 Petaling Jaya, Selangor Darul Ehsan, Malaysia

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FRESENIUS
KABI